



Project Title: 🌱 Household Energy Tracker

Duration: 2-3 Hours

🎯 Objective

As a sustainability analyst, analyze household energy consumption patterns to uncover regional differences and suggest optimizations using NumPy, Pandas, Matplotlib, and Seaborn.

Dataset

household_energy_usage.csv (generate from AI tool)

Dataset Fields:

- Reading_ID
- Household_ID
- Region
- Reading_Date
- Electricity_kWh
- Gas_Consumption_m3
- Water_Liters
- Temperature_C
- Total_Energy_Units

📌 Tasks Checklist

- ◆ Data Preparation

1. Load the dataset and check data types and missing values.
 2. Convert `Reading_Date` into datetime format and extract `Month` and `Day_of_Week`.
 3. Add a new column: `Average_Consumption_Per_Day = Total_Energy_Units / 24`.
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◆ Data Analysis

4. Compare average electricity and gas usage by region.
 5. Identify top 5 households with highest electricity usage.
 6. Determine correlation between `Temperature_C` and `Electricity_kWh`.
 7. Find days with peak total energy consumption.
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◆ Data Visualization

8. Create a line chart of electricity usage over time.
 9. Plot a bar chart comparing gas usage by region.
 10. Use Seaborn to:
 - Create a heatmap of correlations among numeric features.
 - Plot a boxplot of total energy units by region.
 - Create a scatterplot showing temperature vs. electricity usage.
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Submission

- Submit a Python or Jupyter Notebook with all steps, clearly labeled.
- Ensure clean visualizations with proper titles and axis labels.